

Challenges of ec2 using amazon linux

- 1) Launch one **EC2 instance** using **Amazon Linux 2**.

```
#  
~\##### Amazon Linux 2023  
~~\#####\  
~~\###|  
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023  
~~V-' ->  
~~~  
~~~.  
~~~/m/'
```

Last login: Wed Nov 19 09:54:45 2025 from 14.192.14.55
[ec2-user@ip-172-31-36-88 ~]\$ sudo yum update -y
Last metadata expiration check: 20:32:23 ago on Wed Nov 19 09:54:18 2025.
Dependencies resolved.
Nothing to do.
Complete!

2)Install Docker

```
[root@ip-172-31-36-88 ~]# yum install docker
Last metadata expiration check: 20:54:40 ago on Wed Nov 19 09:54:18 2025.
Dependencies resolved.
=====
Package                                Architecture                          Version
=====
Installing:
docker                                 x86_64                               25.0.13-1.amzn2023.0.2
Installing dependencies:
container-selinux                     noarch                               4:2.242.0-1.amzn2023
containerd                             x86_64                              2.1.4-1.amzn2023.0.2
iptables-lib                            x86_64                              1.8.8-3.amzn2023.0.2
iptables-nft                           x86_64                              1.8.8-3.amzn2023.0.2
libcgroup                             x86_64                              3.0-1.amzn2023.0.1
libnetfilter_conntrack                 x86_64                              1.0.8-2.amzn2023.0.2
libnfnetworklink                       x86_64                              1.0.1-19.amzn2023.0.2
libnftnl                               x86_64                              1.2.2-2.amzn2023.0.2
runc                                    x86_64                              1.3.3-2.amzn2023.0.1
Transaction Summary
=====
Install 10 Packages

Total download size: 74 M
Installed size: 280 M
Is this ok [y/N]: y
Downloading Packages:
(1/10): container-selinux-2.242.0-1.amzn2023.noarch.rpm
(2/10): iptables-lib-1.8.8-3.amzn2023.0.2.x86_64.rpm
(3/10): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm
(4/10): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm
(5/10): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm
(6/10): libnfnetworklink-1.0.1-19.amzn2023.0.2.x86_64.rpm
```

- >yum install docker -y
- > systemctl start docker
- > systemctl enable docker

```
[root@ip-172-31-36-88 ~]# systemctl start docker
[root@ip-172-31-36-88 ~]# systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[root@ip-172-31-36-88 ~]# sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Thu 2025-11-20 06:49:55 UTC; 39s ago
 TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 7750 (dockerd)
      Tasks: 9
     Memory: 29.0M
        CPU: 344ms
    CGroup: /system.slice/docker.service
            └─7750 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nfile=32768:65536

Nov 20 06:49:53 ip-172-31-36-88.eu-north-1.compute.internal systemd[1]: Starting docker.service - Docker Application Container Engine.
Nov 20 06:49:53 ip-172-31-36-88.eu-north-1.compute.internal dockerd[7750]: time="2025-11-20T06:49:53.906690184Z" level=info msg="Starting Docker daemon"
Nov 20 06:49:53 ip-172-31-36-88.eu-north-1.compute.internal dockerd[7750]: time="2025-11-20T06:49:53.968068709Z" level=info msg="Loading containers: start."
Nov 20 06:49:53 ip-172-31-36-88.eu-north-1.compute.internal dockerd[7750]: time="2025-11-20T06:49:53.969027204Z" level=info msg="Loading containers: done."
Nov 20 06:49:54 ip-172-31-36-88.eu-north-1.compute.internal dockerd[7750]: time="2025-11-20T06:49:54.986080208Z" level=info msg="Default bridge (docker0) is already up. To create new bridges, use 'docker network create'. If you don't want the bridge, you can remove it with 'docker rm -f docker0'."
Nov 20 06:49:55 ip-172-31-36-88.eu-north-1.compute.internal dockerd[7750]: time="2025-11-20T06:49:55.151614048Z" level=info msg="Loading containers: done."
Nov 20 06:49:55 ip-172-31-36-88.eu-north-1.compute.internal dockerd[7750]: time="2025-11-20T06:49:55.183912127Z" level=info msg="Docker daemon is ready to accept requests."
Nov 20 06:49:55 ip-172-31-36-88.eu-north-1.compute.internal dockerd[7750]: time="2025-11-20T06:49:55.184025467Z" level=info msg="Daemon is ready to accept requests."
```

- >docker — version

```
[root@ip-172-31-36-88 ~]# docker --version
Docker version 25.0.13, build 0bab007
[root@ip-172-31-36-88 ~]#
```

3)Install Jenkins

STEP 1 — Install Java (mandatory for Jenkins)

- > sudo yum (or) dnf install java-17-amazon-corretto -y will install java packages

```
[root@ip-172-31-36-88 ~]# sudo dnf install java-17-amazon-corretto -y
Last metadata expiration check: 21:17:08 ago on Wed Nov 19 09:54:18 2025.
Dependencies resolved.
=====
Package                                         Architecture                               Version
=====
Installing:
 java-17-amazon-corretto                       x86_64                                    1:17.0.1
Installing dependencies:
 alsa-lib                                       x86_64                                    1.2.7.2-5
 cairo                                          x86_64                                    1.18.0-4
 dejavu-sans-fonts                             noarch                                    2.37-16.1
 dejavu-sans-mono-fonts                       noarch                                    2.37-16.1
 dejavu-serif-fonts                           noarch                                    2.37-16.1
 fontconfig                                     x86_64                                    2.13.94-2
 fonts-filesystem                             noarch                                    1:2.0.5-3
 freetype                                       x86_64                                    2.13.2-5
```

—> with that packages we can install Jenkins

—> now

STEP 2 — Add the official Jenkins repository

By using wget command and paste that Jenkins repo

It will download Jenkins repo

```
[root@ip-172-31-36-88 ~]# sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo
sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
--2025-11-20 07:13:16-- https://pkg.jenkins.io/redhat-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 199.232.174.133, 2a04:4e42:6b::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|199.232.174.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 85
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo      100%[=====]
2025-11-20 07:13:16 (5.01 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [85/85]

[root@ip-172-31-36-88 ~]# yum update -y
Jenkins-stable
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-36-88 ~]#
```

CloudShell Feedback Console Mobile App

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—> now we can install Jenkins package

—> yum install jenkins

```
Complete!
[root@ip-172-31-36-88 ~]# yum install jenkins
Last metadata expiration check: 0:00:52 ago on Thu Nov 20 07:13:52 2025.
No match for argument: jenkins
Error: Unable to find a match: jenkins
[root@ip-172-31-36-88 ~]# yum install jenkins -y
Last metadata expiration check: 0:01:14 ago on Thu Nov 20 07:13:52 2025.
Dependencies resolved.

=====
Package                                Architecture                               Version
=====
Installing:
jenkins                                noarch                                     2.528.2-1.1.noarch

Transaction Summary
=====
Install 1 Package

Total download size: 91 M
Installed size: 91 M
Downloading Packages:
jenkins-2.528.2-1.1.noarch.rpm
=====

Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :
  Running scriptlet: jenkins-2.528.2-1.1.noarch
  Installing     : jenkins-2.528.2-1.1.noarch
  Running scriptlet: jenkins-2.528.2-1.1.noarch
  Verifying      : jenkins-2.528.2-1.1.noarch

Installed:
jenkins-2.528.2-1.1.noarch

Complete!
[root@ip-172-31-36-88 ~]#
```

```
[root@ip-172-31-36-88 ~]# vi /usr/lib/systemd/system/jenkins.service
[root@ip-172-31-36-88 ~]#
```

Now for editing the port go inside jenkins.service

And change the port number

```
# IP address to listen on for all requests.
# The default is to listen on all interfaces (0.0.0.0).
#Environment="JENKINS_LISTEN_ADDRESS="

# Port to listen on for HTTP requests. Set to -1 to disable.
# To be able to listen on privileged ports (port numbers less than 1024),
# add the CAP_NET_BIND_SERVICE capability to the AmbientCapabilities
# directive below.
Environment="JENKINS_PORT=8081"

# IP address to listen on for HTTPS requests. Default is disabled.
#Environment="JENKINS_HTTPS_LISTEN_ADDRESS="

# Port to listen on for HTTPS requests. Default is disabled.
# To be able to listen on privileged ports (port numbers less than 1024),
# add the CAP_NET_BIND_SERVICE capability to the AmbientCapabilities
# directive below.
#Environment="JENKINS_HTTPS_PORT=443"

# Path to the keystore in JKS format (as created by the JDK's keytool).
# Default is disabled.
#Environment="JENKINS_HTTPS_KEYSTORE=/path/to/keystore.jks"

# Password to access the keystore defined in JENKINS_HTTPS_KEYSTORE.
# Default is disabled.
```

We have changed the port to 8081

—> now systemctl daemon-reload

—> now start

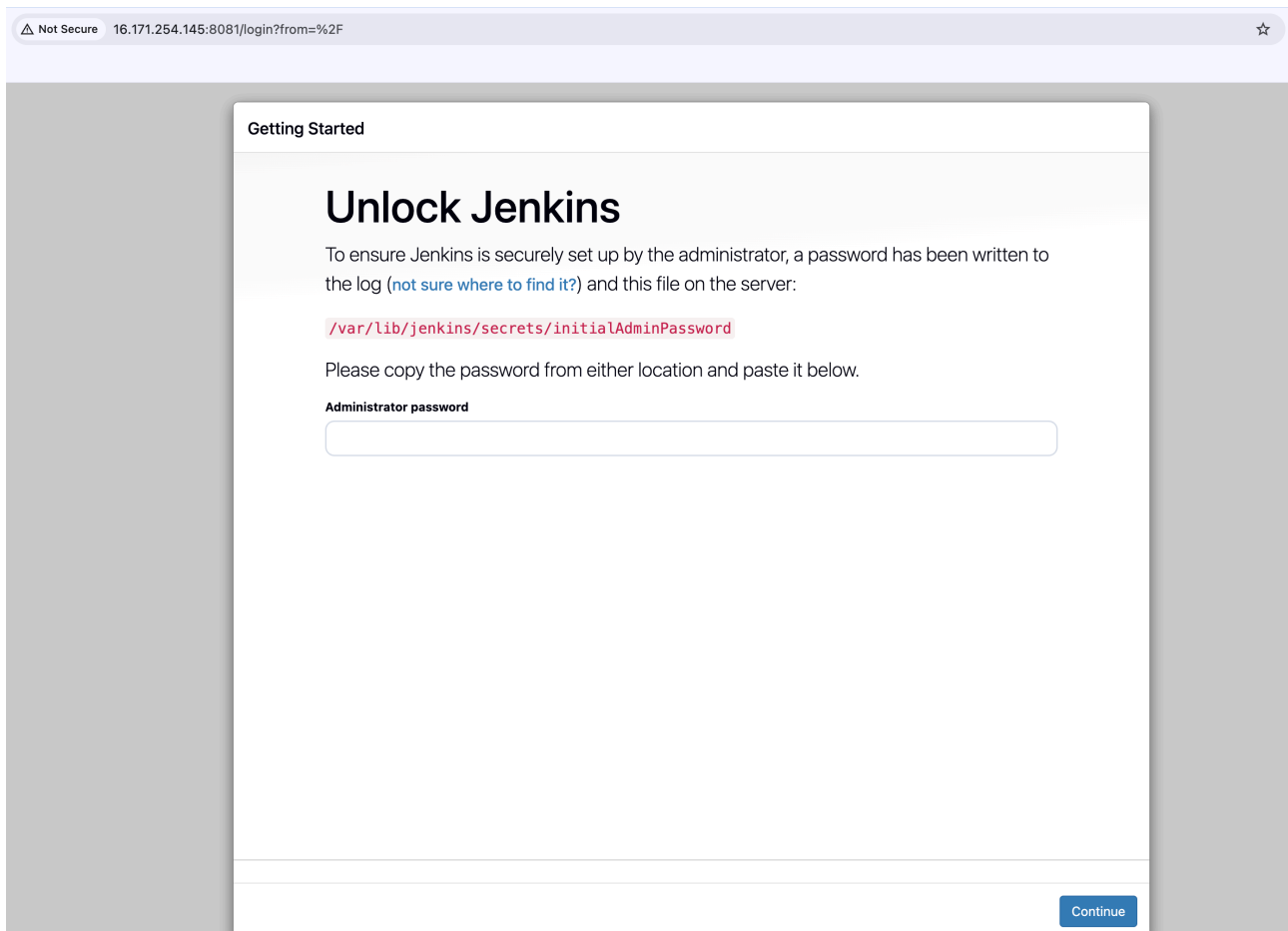
—> systemctl start jenkins

—> systemctl status jenkins

```
[root@ip-172-31-36-88 ~]# C
[root@ip-172-31-36-88 ~]# sudo systemctl daemon-reload
[root@ip-172-31-36-88 ~]# systemctl start jenkins
[root@ip-172-31-36-88 ~]# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; disabled; preset: disabled)
   Active: active (running) since Thu 2025-11-20 07:21:00 UTC; 40s ago
 Main PID: 10293 (java)
    Tasks: 43 (limit: 1053)
  Memory: 351.3M
     CPU: 20.752s
    CGroup: /system.slice/jenkins.service
            └─10293 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --w

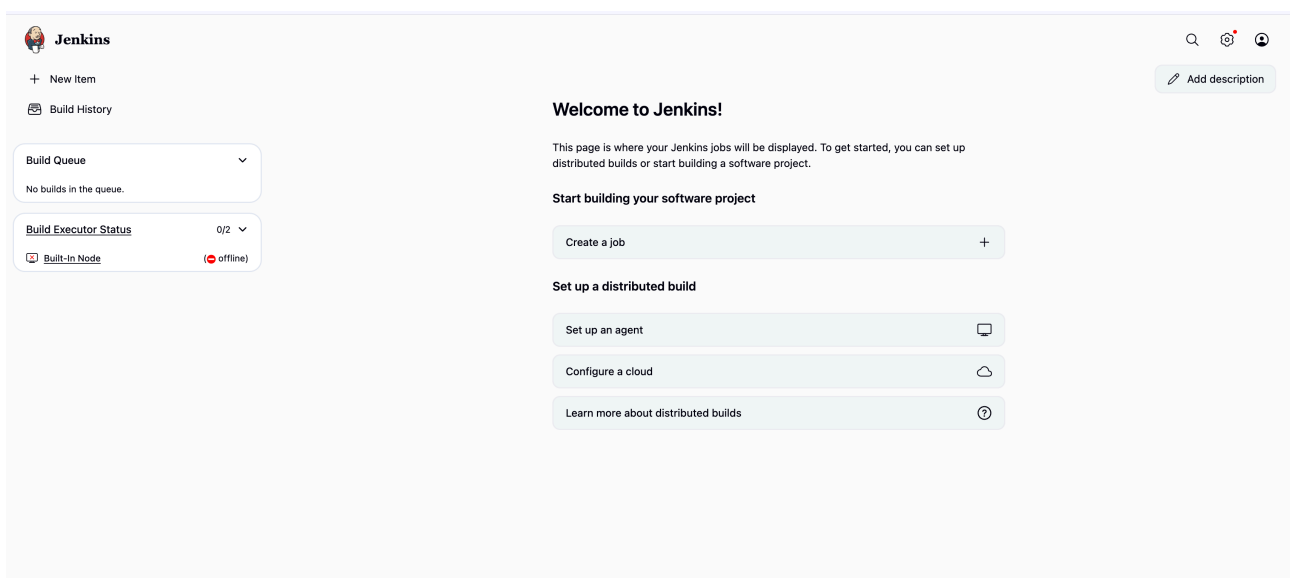
Nov 20 07:20:54 ip-172-31-36-88.eu-north-1.compute.internal jenkins[10293]: [LF]>
Nov 20 07:20:54 ip-172-31-36-88.eu-north-1.compute.internal jenkins[10293]: [LF]> *****
Nov 20 07:20:54 ip-172-31-36-88.eu-north-1.compute.internal jenkins[10293]: [LF]> *****
Nov 20 07:20:54 ip-172-31-36-88.eu-north-1.compute.internal jenkins[10293]: [LF]> *****
Nov 20 07:21:00 ip-172-31-36-88.eu-north-1.compute.internal jenkins[10293]: 2025-11-20 07:21:00.
Nov 20 07:21:00 ip-172-31-36-88.eu-north-1.compute.internal jenkins[10293]: 2025-11-20 07:21:00.
Nov 20 07:21:00 ip-172-31-36-88.eu-north-1.compute.internal systemd[1]: Started jenkins.service
Nov 20 07:21:00 ip-172-31-36-88.eu-north-1.compute.internal jenkins[10293]: 2025-11-20 07:21:00.
Nov 20 07:21:00 ip-172-31-36-88.eu-north-1.compute.internal jenkins[10293]: 2025-11-20 07:21:00.
Nov 20 07:21:05 ip-172-31-36-88.eu-north-1.compute.internal jenkins[10293]: 2025-11-20 07:21:05.
lines 1-20/20 (END)
```

**Copy public ip and add port 8081 it will open the page
—> if the page is not opening then add inbound rule for port 8081 in security group**



We can cat the password in —> initialadminpassword

```
[root@ip-172-31-36-88 ~]# sudo cat /var/lib/jenkins/secrets/initialAdminPassword
f28faa33456245cb9333b3adc968e747
[root@ip-172-31-36-88 ~]#
```



4) Install Apache

—> httpd is active running in port 80

```
[root@ip-172-31-36-88 ~]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Thu 2025-11-20 06:23:46 UTC; 28min ago
     Docs: man:httpd.service(8)
  Main PID: 1444 (httpd)
    Status: "Total requests: 1; Idle/Busy workers 100/0; Requests/sec: 0.000584; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1053)
    Memory: 17.9M
       CPU: 1.408s
    CGroup: /system.slice/httpd.service
            └─1444 /usr/sbin/httpd -DFOREGROUND
              └─1445 /usr/sbin/httpd -DFOREGROUND
                └─1446 /usr/sbin/httpd -DFOREGROUND
                  └─1448 /usr/sbin/httpd -DFOREGROUND
                    └─1467 /usr/sbin/httpd -DFOREGROUND

Nov 20 06:23:46 ip-172-31-36-88.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
```



—> vi /var/www/html/index.html

It will open deployment location of httpd

```
<h1>Amazon Linux 2 - Apache from user-data</h1>
```

5) Install Nginx. -> yum install nginx -y

```
[root@ip-172-31-36-88 html]# vi index.html
[root@ip-172-31-36-88 html]# yum install nginx -y
Last metadata expiration check: 21:03:56 ago on Wed Nov 19 09:54:18 2025.
Dependencies resolved.
=====
Package                                Architecture                               Version
=====
Installing:
nginx                                  x86_64                                     1:1.28.0-1.amzn2023.0.2
Installing dependencies:
gperftools-libs                       x86_64                                     2.9.1-1.amzn2023.0.3
libunwind                             x86_64                                     1.4.0-5.amzn2023.0.3
nginx-core                             x86_64                                     1:1.28.0-1.amzn2023.0.2
nginxfilesystem                       noarch                                     1:1.28.0-1.amzn2023.0.2
nginx-mimetypes                       noarch                                     2.1.49-3.amzn2023.0.3
=====
Transaction Summary
=====
Install 6 Packages

Total download size: 1.1 M
Installed size: 3.6 M
Downloading Packages:
(1/6): nginx-1.28.0-1.amzn2023.0.2.x86_64.rpm
(2/6): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm
(3/6): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm
(4/6): nginxfilesystem-1.28.0-1.amzn2023.0.2.noarch.rpm
(5/6): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm
(6/6): nginx-core-1.28.0-1.amzn2023.0.2.x86_64.rpm
=====
Total
Running transaction check
```

```
user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

# Load dynamic modules. See /usr/share/doc/nginx/README.dynamic.
include /usr/share/nginx/modules/*.conf;

events {
    worker_connections 1024;
}

http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile on;
    tcp_nopush on;
    keepalive_timeout 65;
    types_hash_max_size 4096;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    # Load modular configuration files from the /etc/nginx/conf.d directory.
    # See http://nginx.org/en/docs/nginx_core_module.html#include
    # for more information.
    include /etc/nginx/conf.d/*.conf;

    server {
        listen 81;
        listen [::]:80;
        server_name _;
        root /usr/share/nginx/html;

        # Load configuration files for the default server block.
        include /etc/nginx/default.d/*.conf;
```

For changing port open
—> **vi /etc/nginx/nginx.conf**



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Running the nginx on port 81 in same ec2

6) Install Apache Tomcat.

—> for installing tomcat

—> need java

After installing java

—> use wget and past the tomcat app url and download its tar cvf of tomcat

—> later extract that tar file using

—> tar xvf

—> now go inside of tomcat directory

```
Last login: Thu Nov 20 08:06:41 2025 from 14.192.14.61
[ec2-user@ip-172-31-36-88 ~]$ ls
apache-tomcat-9.0.112.tar.gz  tomcat
[ec2-user@ip-172-31-36-88 ~]$ cd tomcat
[ec2-user@ip-172-31-36-88 tomcat]$ ls
```

—> in side bin directory we have our startup.sh script to start tomcat

Tomcat/bin/startup.sh

```
[ec2-user@ip-172-31-36-88 tomcat]$ cd bin
[ec2-user@ip-172-31-36-88 bin]$ ./startup.sh
Using CATALINA_BASE:   /home/ec2-user/tomcat
Using CATALINA_HOME:   /home/ec2-user/tomcat
Using CATALINA_TMPDIR: /home/ec2-user/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /home/ec2-user/tomcat/bin/bootstrap.jar:/home/ec2-user/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[ec2-user@ip-172-31-36-88 bin]$
```


For changing port we have to go inside of server.xml file

—> tomcat/conf/server.xml

—> we have changed port to 9090

```
<!--
<Executor name="tomcatThreadPool" namePrefix="catalina-exec-"
      maxThreads="150" minSpareThreads="4"/>
-->

<!-- A "Connector" represents an endpoint by which requests are received
and responses are returned. Documentation at :
Java HTTP Connector: /docs/config/http.html
Java AJP  Connector: /docs/config/ajp.html
APR (HTTP/AJP) Connector: /docs/apr.html
Define a non-SSL/TLS HTTP/1.1 Connector on port 8080
-->
<Connector port="9090" protocol="HTTP/1.1"
      connectionTimeout="20000"
      redirectPort="8443"
      maxParameterCount="1000"
      />
<!-- A "Connector" using the shared thread pool-->
<!--
<Connector executor="tomcatThreadPool"
      port="8080" protocol="HTTP/1.1"
      connectionTimeout="20000"
      redirectPort="8443"
      maxParameterCount="1000"
      />
-->
<!-- Define an SSL/TLS HTTP/1.1 Connector on port 8443
This connector uses the NIO implementation. The default
SSLImplementation will depend on the presence of the APR/native
library and the useOpenSSL attribute of the AprLifecycleListener.
Either JSSE or OpenSSL style configuration may be used regardless of
the SSLImplementation selected. JSSE style configuration is used below.
-->
<!--
<Connector port="8443" protocol="org.apache.coyote.http11.Http11NioProtocol"
      maxThreads="150" SSLEnabled="true"
      maxParameterCount="1000"
-- INSERT --
```

—>16.171.42.99:9090 (port 9090) tomcat

△ Not Secure 16.171.42.99:9090

Home Documentation Configuration Examples Wiki Mailing Lists

Apache Tomcat/9.0.112

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Developer Quick Start

- [Tomcat Setup](#)
- [First Web Application](#)
- [Realms & AAA](#)
- [JDBC DataSources](#)
- [Examples](#)
- [Servlet S](#)
- [Tomcat V](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 9.0 access to the manager application is split between different users. [Read more...](#)

[Release Notes](#)

[Changelog](#)

[Migration Guide](#)

[Security Notices](#)

Documentation

[Tomcat 9.0 Documentation](#)

[Tomcat 9.0 Configuration](#)

[Tomcat Wiki](#)

Find additional important configuration information in:

`$CATALINA_HOME/RUNNING.txt`

Developers may be interested in:

- [Tomcat 9.0 Bug Database](#)
- [Tomcat 9.0 JavaDocs](#)
- [Tomcat 9.0 Git Repository at GitHub](#)

Getting Help

[FAQ and Maili](#)

The following ma

- [tomcat-announce](#)
Important announ
vulnerability notif
- [tomcat-users](#)
User support and d
- [taglibs-user](#)
User support and d
- [tomcat-dev](#)
Development mailin
messages

Other Downloads Other Documentation Get Involved Miscellaneous

—>16.171.42.99:8081 (port 8081) jenkins

△ Not Secure 16.171.42.99:8081/login?from=%2F ☆



Sign in to Jenkins

Username

Password

☐ Keep me signed in

[Sign in](#)

—>16.171.42.99:81 (port 81) nginx



16.171.42.99:80 (port 80) httpd



Amazon Linux 2 - Apache from user-data

So

—

—> httpd on port 80 16.171.42.99.80

—> nginx on port 81

—> jenkins on port 8081

—> tomcat on port 9090

In same ec2 server

